

SIMATS ENGINEERING ASSESSMENT TOOL 3

COURSE NAME: COMPUTER NETWORKS
COURSE CODE: CSA07

COURSE OUTCOME COVERED

CO4:

Design network topologies star, mesh, bus, and hybrid using networking devices, transmission media, and cabling standards
(BL2,BL6)

Assessment Tool Weightage: 30%

QUESTIONS: 5
Marks:25

Total

Annexure A

Comparative Analysis

Code of Conduct:

I T.Ganesh(192425246) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

T.Ganesh

Signature of the student:

Annexure A

Comparative Analysis

1. Star topology is generally more expensive than bus topology because it requires a central device such as a switch or hub and more cabling to connect each node individually. In contrast, bus topology uses a single backbone cable, making it low-cost and easier to install. However, in terms of reliability, star topology is superior because failure of one node does not affect others, whereas a failure in the main bus cable can bring down the entire network. Performance is also better in star topology since data transmission is managed through a central device, reducing collisions, while bus topology suffers from performance degradation as more devices are added due to increased data traffic on the shared medium.

Solution 1: Star vs Bus Topology — Cost, Reliability, Performance

Parameter	Star Topology	Bus Topology
Cost	Higher — needs central switch/hub + more cabling	Lower — single shared backbone cable
Reliability	High — one node failing does not affect others	Low — backbone failure disables entire network
Performance	Better — central device manages traffic, fewer collisions	Degrades as more devices are added (shared medium)
Verdict	Preferred for reliability & performance despite higher cost	Preferred only for very low-cost, small setups

□ **Cost & Cabling:**

- **Star Topology:** Higher cost; requires a central device (hub/switch) and dedicated cabling for each node.
- **Bus Topology:** Lower cost; uses a single shared backbone cable, making it cheaper and easier to install.

□ **Reliability:**

- **Star Topology:** High; an individual node failure does not affect the rest of the network.
- **Bus Topology:** Low; a break or fault in the main backbone cable brings down the entire network.

□ **Performance:**

- **Star Topology:** High; central device manages traffic, significantly reducing collisions.
- **Bus Topology:** Degrades under heavy load; adding more devices increases data traffic and collisions on the shared medium.

2. Mesh topology offers very high fault tolerance because every node is connected to multiple other nodes, providing multiple paths for data transmission. If one link fails, data can still be routed through alternate paths without disrupting communication. In contrast, star topology has moderate fault tolerance; while failure of individual nodes does not affect the network, failure of the central hub or switch can cause the entire network to stop functioning. Therefore, mesh topology is more robust and reliable in critical environments compared to star topology.

Solution 2: Mesh vs Star Topology — Fault Tolerance

Parameter	Mesh Topology	Star Topology
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Redundant Paths	Yes — every node linked to multiple nodes	No — all traffic passes through central hub
Fault Tolerance	Very high — alternate paths bypass failed links	Moderate — single node failure isolated, but hub failure is fatal
Critical-use Suitability	Best suited for mission-critical networks	Suitable for general-purpose, cost-sensitive networks
Verdict	More robust and reliable overall	Simpler but has a single point of failure (the hub)

3. Bus topology is simple and easy to install because it uses a single backbone cable with minimal connections, making it suitable for small networks. On the other hand, mesh topology is highly complex to install as each node must be connected to every other node, requiring a large number of cables and ports. This complexity increases significantly as the number of devices grows, making mesh topology time-consuming and costly to implement compared to the straightforward setup of bus topology.

Solution 3: Bus vs Mesh Topology — Installation Complexity

Parameter	Bus Topology	Mesh Topology
Cabling Required	Minimal — single backbone cable	Very high — $n(n-1)/2$ links for full mesh
Installation Effort	Simple and quick	Complex, time-consuming, needs many ports
Scalability of Install Effort	Grows slowly with devices	Grows exponentially with devices
Verdict	Best for small, simple networks	Best only where full redundancy justifies the cost/effort

4. Hybrid topology is highly scalable because it combines two or more different topologies (such as star, mesh, or bus), allowing the network to expand flexibly according to organizational needs. New segments can be added without affecting the existing

network structure. In comparison, star topology is also scalable but to a limited extent, as it depends on the capacity of the central device (switch or hub). Once the central device reaches its port limit, expansion requires additional hardware. Thus, hybrid topology provides greater scalability than star topology.

Solution 4: Hybrid vs Star Topology — Scalability

Parameter	Hybrid Topology	Star Topology
Expansion Method	Combine multiple topologies as needed	Add nodes until central device port limit is reached
Limiting Factor	Minimal — flexible by design	Central switch/hub capacity
Impact on Existing Network	New segments added without disruption	Expansion may need new hardware
Verdict	Greater long-term scalability	Limited, moderate scalability

5. In terms of maintenance, star topology is relatively easy to manage because issues can be quickly identified and isolated to a specific node or cable. Mesh topology, while highly reliable, is difficult to maintain due to its complex structure and large number of connections, making troubleshooting time-consuming. Bus topology is harder to maintain than star because faults in the backbone cable can disrupt the entire network and are difficult to locate. Hybrid topology has moderate maintenance complexity; while it benefits from flexibility, managing multiple combined topologies requires careful monitoring and skilled administration.

Solution 5: Maintenance Comparison Across Topologies

Topology	Maintenance Characteristic
Star	Easy — faults isolated to a specific node/cable
Mesh	Difficult — complex structure, many connections to check
Bus	Hard — backbone faults affect whole network and are hard to locate

Hybrid	Moderate — flexible but needs skilled, careful administration
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RUBRICS FOR CALCULATION

Criteria	Weight age	Exemplary (4)	Proficient (3)	Developing (2)	Beginning (1)
Understanding of Concepts	15%	Demonstrates complete and in-depth understanding of all concepts being compared	Good understanding with minor gaps	Basic understanding; some concepts unclear	Poor understanding of concepts
Comparison Depth	20%	Provides detailed, insightful, and meaningful comparisons across multiple aspects	Adequate comparison with relevant points	Limited comparison; lacks depth	Minimal or incorrect comparison
Criteria Selection	10%	Selects highly relevant and appropriate comparison parameters	Mostly relevant criteria	Some irrelevant or missing criteria	Poor or no criteria selection
Analytical Skills	15%	Strong analysis with logical	Good analysis with some	Basic analysis	No clear analysis

Criteria	Weight age	Exemplary (4)	Proficient (3)	Developing (2)	Beginning (1)
		reasoning and clear justification	justificatio n	with weak reasoning	
Organizatio n & Structure	10%	Well- organized, clear, and logically structured	Generally organized	Somewhat disorganize d	Poorly structured
Use of Examples / Evidence	10%	Uses strong, relevant examples or data to support comparison	Uses some examples	Limited examples	No supporting examples
Clarity of Presentatio n	10%	Very clear, concise, and easy to understand	Mostly clear	Somewhat unclear	Difficult to understand
Conclusion & Insights	10%	Insightful conclusion with strong justification	Clear conclusion	Basic conclusion	No or weak conclusion